Enhance Ground Proximity Warning System (TAWS) (Mod 6/2049) – General Data

1. General

The Enhanced Ground Proximity Warning System (EGPWS) (TAWS) provides alerts and warnings to prevent controlled flight into terrain. This is part of the Public Commercial Operation Bundle customer option order 6210.

For aircraft with S.O.O. 6388 incorporated, there is no Enhanced Ground Proximity Warning System (EGPWS) (TAWS) installed on the aircraft.

For aircraft with S.O.O. 6388 incorporated, the Enhanced Ground Proximity Warning System (EGPWS) (TAWS) and associated systems are installed on the aircraft by incorporating customer Standard Order Option S.O.O. 6387.

2. Description

A. Enhanced Ground Proximity Warning System (EGPWS)

The Honeywell MK VI Enhanced Ground Proximity Warning System (EGPWS) consists of a TAWS computer installed in the nose avionics bay and a configuration module mounted as part of the P2 connector backshell. TAWS provides an enhanced capability of reducing accidents cause by controlled flight into terrain. The system achieves this by receiving a variety of aircraft parameters as inputs, then applying alerting algorithms to provide the flight crew with aural messages and visual annunciations and display.

The Terrain Avoidance Warning System (TAWS) unit receives GPS position via the MAU for accurate position determination in conjunction with a regional database. One of three databases may be installed in the system (Atlantic, Americas or Pacific). The databases also contain the locations of all runways longer than 2000 feet (600 m) that have a published instrument approach. A visual depiction of the terrain proximate to the aircraft can be displayed on the PFD and the MFD.

The terrain alerting algorithms continuously compute the terrain clearance envelopes ahead of the aircraft. If the boundaries of these envelopes conflict with terrain elevation data in the terrain database, then alerts are issued. Two envelopes are computed, one corresponding to a terrain caution alert and the other to a terrain warning alert. Terrain awareness caution and warning alerts are inhibited below 30 feet of radio altitude, within 1 nautical mile of the runway, and below 30 knots groundspeed.

When the conditions exist to generate a terrain or obstacle caution alert, the terrain image on the PFD TAWS Overlay is enhanced to highlight the threatening terrain as solid yellow for caution threats and the appropriate aural alert is given. When the required conditions have been met to generate a terrain or obstacle warning alert, the display image on the PFD TAWS Overlay is enhanced to highlight the terrain as solid red and the appropriate aural alert is given.

The TAWS sends aural alert messages, when necessary, to the audio control panel and from there onward to the headsets. At the same time annunciations are displayed on the PFD ADI in an amber box for GND PROX or red for PULL UP. The annunciations flash in reverse video for 5 seconds and then remain on until the condition is no longer detected. If the terrain overlay is not selected for display and a TAWS alert is generated, the terrain overlay will be displayed (automatic pop-up) on the HSI in the partial compass mode. The TAWS sends a suppression signal to the TCAS to inhibit voice messages from the TCAS when the TAWS is generating voice messages.

28 Volt DC power is protected by a 5 Amp TAWS (Left Bus) circuit breaker located on the avionics circuit breaker panel (refer to 24-00-00, Electrical Power – General Data).

The EGPWS (MK VI) interfaces to the following aircraft systems:

1) Modular Avionics Unit (MAU) (refer to 31-43-00, Modular Avionics Unit (MAU) – General Data).

2) Global Positioning System (GPS) (refer to 34-56-00, Global Positioning System (GPS) (Mod 6/2049) – General Data).

3) Air Data, Attitude and Heading Reference System (ADAHRS) (refer to 34-23-00, Air Data, Attitude and Heading Reference System (ADAHRS) (KSG 7200) – General Data).

4) Traffic Alert and Collision Avoidance System (TCAS) (refer to 34-43-00, Traffic Alert and Collision Avoidance System I (TCAS I) – General Data and 34-43-01, Traffic Alert and Collision Avoidance System II (TCAS II) (Mod S.O.O. 6219) – General Data), if installed.

5) Radar Altimeter System (refer to 34-42-00, Radar Altimeter System (Mod 6/2049) – General Data).

3. Adjustment/Test

A. Function Test EGPWS (MK VI) – (TAWS)

Note The aircraft must be in a position to receive signals from the GPS satellites prior to doing the following function tests.

1) General

These procedures are for Levels 1 to 6 of Self Test (ST) that describe the current condition and configuration of the EGPWS (MK VI) – (TAWS).

2) Function Test

a) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

b) Wait for the aircraft electrical and avionics systems to initialize.

c) Ensure that TAWS (G5) (Left Bus) circuit breaker on the avionics circuit breaker panel is closed and the EGPWS Computer has finished its boot sequence.

d) Ensure that the TERRAIN INHIB message is not displayed on the HSI Indicator. If displayed, deactivate Terrain Inhibit via the TERR INHIB switch on the MF Controller.

e) Push the soft key self test button for the TAWS, which is found on MFD 2 display in the Systems Information Section (lower left of screen), following the sequence below:

1 Use the 'Arrow Keys' to navigate the selection over the 'Systems Info' section of the MFD 2.

2 Push the 'Page' key on the MF Controller.

3 Select 'WX/LX/TAWS' from the drop down menu.

4 Push the 'Enter' key on the MF Controller.

5 Use the Joystick on the MF Controller to select the 'TAWS' tab.

6 Push the 'Enter' key on the MF Controller.

7 Use the Joystick on the MF Controller to select the 'TAWS SELF TEST'.

8 Push the 'Enter' key on the MF Controller.

f) Verify the following Level 1 test sequence:

- The red PULL UP annunciator is displayed on the PFDs.
- Aural message EGPWS SYSTEM OK is heard.
- The PFDs show a terrain test pattern sweep.
- The red PULL UP annunciator turns off.
- Amber GND PROX annunciator turns off.
- The PFD terrain test pattern turns off.

- Amber GND PROX annunciator turns off.

g) Within 2 seconds, select the TAWS SELF TEST softkey to continue to Level 2.

h) Verify no faults for Level 2.

i) When ‘press to continue' is heard, select the TAWS SELF TEST softkey within 2 seconds to continue to Level 3.

j) Verify correct System Configuration for Level 3.

k) When ‘press to continue' is heard, select the TAWS SELF TEST softkey within 2 seconds to continue to Level 4.

l) Level 4 lists Fault History record.

m) When ‘press to continue' is heard, select the TAWS SELF TEST softkey within 2 seconds to continue to Level 5.

n) Level 5 lists Warning History record of flights.

o) When ‘press to continue' is heard, select the TAWS SELF TEST softkey within 2 seconds to continue to Level 6.

p) Level 6 initiates a Discrete Input Test. Voice audio is annunciated when the state of a discrete input is changed.

q) Select the TAWS SELF TEST sofkey to disable the Discrete Input Test.

r) Verify ‘End of Self Test' is annunciated.

s) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

B. Loading a Database onto the EGPWS MK VI – (TAWS)

1) General

These procedures are for loading a Terrain Database onto the EGPWS Computer. The EGPWS Computer has a database preloaded from factory. Verification of the Terrain Database should be done first, to confirm the desired database is not already installed.

The MK VI EGPWS, 965–1180–0XX, is shipped from the factory loaded with the Regional Terrain Database, requested when the purchase order was placed. Use (operation) of a MK VI EGPWS outside of the loaded Regional Terrain Database will result in the Terrain Awareness function being unavailable. Aircraft operating outside the loaded Regional Terrain Database region will have to load one of the other two Regional Terrain Databases before beginning the ground test.

The EGPWS Computer must be powered up and must have finished its boot sequence. Access to the EGPWS computer, located on the lower right avionics shelf, is required. The Smart Cable (Honeywell PN 951–0386–001) and a PCMCIA data storage card with the desired database on it is required.

2) Function Test

Note Precautionary notes on the PCMCIA card, regarding insertion and/or removal while power is applied, should be ignored since the EGPWC automatically handles the application and removal of PCMCIA card power.

a) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

b) Wait for the aircraft electrical and avionics systems to initialize.

c) Ensure that the TAWS (G5) circuit breaker on the left bus is pushed in and the EGPWS Computer has finished its boot sequence.

d) Connect the Smart Cable (Honeywell PN 951–0386–001) to EGPWC J3 connector.

e) Insert the PCMCIA card into the Smart Cable PCMCIA card slot.

f) While the loading is in progress the IN PROG LED on the Smart Card remains ON and the COMPUTER OK LED on the EGPWC is OFF.

g) When loading is complete the XFER COMP LED on the Smart Card turns ON.

h) Remove the PCMCIA card from the Smart Card slot.

i) After approximately 15 seconds the COMPUTER OK LED comes ON to indicated that the contents of the PCMCIA card were successfully transferred.

j) Remove the Smart Card connector from the EGPWC front panel J3 connector.

k) Verify the Terrain Database Version to make sure the new Database has been loaded correctly.

l) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

C. Verification of Terrain Database EGPWS (MK VI) – (TAWS)

Note The aircraft must be in a position to receive signals from the GPS satellites prior to doing the following function test.

1) General

These procedures are for the verification of the Terrain Database loaded into the EGPWS (MK VI) – (TAWS).

2) Function Test

a) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

b) Wait for the aircraft electrical and avionics systems to initialize.

c) Ensure that TAWS (G5) (Left Bus) circuit breaker on the avionics circuit breaker panel is closed and that the COMPUTER OK LED on the EGPWC front panel is on.

d) Ensure that the TERRAIN INHIBIT message is not displayed on the HSI Indicator. If displayed, deactivate Terrain Inhibit via the TERR INHIB switch on the MF Controller.

e) Select the TAWS from the pull down menu, which is activated via the OVRLY soft key located on the outboard side on either PFD.

f) Select a Range on the HSI display between 300 and 2.5 miles.

g) Push the soft key self test button for the TAWS, which is found on MFD 2 display in the Systems Information Section (lower left of screen), following the sequence below:

1 Use the 'Arrow Keys' to navigate the selection over the 'Systems Info' section of the MFD 2.

2 Push the 'Page' key on the MF Controller.

3 Select 'WX/LX/TAWS' from the drop down menu.

4 Push the 'Enter' key on the MF Controller.

5 Use the Joystick on the MF Controller to select the ‘TAWS' tab.

6 Push the ‘Enter' key on the MF Controller.

7 Use the Joystick on the MF Controller to select the ‘TAWS SELF TEST'.

8 Push the ‘Enter' key on the MF Controller.

h) Verify the new Database Version, which is displayed on the test patter on the HSI Indicator on either PFD.

i) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

D. Configuration Module Programming

1) General

These procedures are provided to program the configuration module.

2) Program the Configuration Module

a) Gain access to the forward avionic shelves through the forward baggage compartment.

b) Use a D-sub cable to connect a personal computer (PC) with WinViews software installed to the maintenance connector (J3) on the front of the EGPWS unit.

c) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

d) Wait for the aircraft electrical and avionics systems to initialize.

e) Ensure that TAWS (G5) (Left Bus) circuit breaker on the avionics circuit breaker panel is closed.

f) Start the WinViews software on the PC.

g) Select COM 1 and then CONNECT. Ensure that the Terminal Mode Page is shown.

h) At the C: prompt type CFG and then press ENTER.

i) Type the following text:

CUW 0/14 3 5 4 1 2 9 93 2 3 5 2 0 1 2/

j) Press ENTER.

k) The cursor will flash with Y or N displayed.

l) Type Y. The EGPWS unit will reboot. Ensure this is complete before continuing.

m) Type CTRL Z.

n) Type PS and press ENTER.

o) Ensure that the correct configuration has been programmed. Ensure the display reads:

CUW 0/14 3 5 4 1 2 9 93 2 3 5 2 0 1 2/

p) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

q) Remove the D-sub cable connected to the maintenance connector (J3) on the front of the EGPWS unit and turn off the PC.

4. Removal/Installation

WARNING **INSTALLATION OF THIS COMPONENT CONTAINS PROTECTION FEATURES THAT ARE CRITICAL TO PREVENTING DEGRADATION FROM THE EFFECTS OF HIGH INTENSITY RADIATED FIELDS (HIRF) AND INDIRECT EFFECT OF LIGHTNING (IEL). ENSURE INTEGRITY OF THE HIRF/IEL PROTECTION FEATURES ARE MAINTAINED IN ACCORDANCE WITH THE FOLLOWING:**

CHASSIS GROUNDING – ALL RACK MOUNT / REMOTE MOUNT UNITS SHALL BE ELECTRICALLY BONDED TO THE AIRFRAME. ENSURE THAT THE MATING SURFACES BETWEEN THE LRU MOUNTING TRAY (OR LRU MOUNTING FEET IF A TRAY IS NOT USED) PROVIDE A LOW IMPEDANCE ELECTRICAL PATH. ALL MATING SURFACES MUST BE FREE OF ALL PAINT AND OTHER NONCONDUCTIVE ELEMENTS. REFER TO BONDING REQUIREMENTS IN 20-75-00, ELECTRICAL BONDING AND GROUNDING OF AIRCRAFT STRUCTURES - GENERAL DATA).

SHIELDS GROUNDS – VERIFY SHIELD GROUNDING AT CONNECTOR BACKSHELLS AS PER REQUIREMENTS IN PSM 1-64-2W.

BACKSHELLS OR STRAIN RELIEFS – VERIFY PROPER ELECTRICAL BONDING BETWEEN THE BACKSHELLS/STRAIN RELIEF AND THE MOUNTING TRAY OR LRU CASE AS PER REQUIREMENTS IN PSM 1-64-2W.

A. Removal and Installation of EGPWS (MK VI) – (TAWS)

1) General

These procedures are for the removal and installation of the EGPWS (MK VI) – (TAWS).

2) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Pull circuit breaker TAWS (G5) (Left Bus) on the avionics circuit breaker panel and fit safety clip.

c) Gain access to the forward avionic shelves through the forward baggage compartment.

d) Disconnect the electrical connector on the radar EGPWS unit.

e) Loosen retaining screw clamp from the front and slide the unit out of the mounting tray.

f) Visually inspect mounting tray for condition, if damaged replace as per Para 4.B.

g) Remove unit from aircraft.

Note Stow connectors of the removed components.

3) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure the TAWS (G5) (Left Bus) circuit breaker is pulled and fitted with safety clip.

c) Slide the unit onto the mounting tray. Tighten the retaining screw clamp at the front lip of the EGPWS unit.

d) Reconnect the electrical connector to the EGPWS unit. Ensure connections are secure.

e) Install top and middle covers attached to bulkhead station 37.21.

WARNING **ACCESS COVERS ARE REQUIRED TO MAINTAIN PROTECTION FEATURES THAT ARE ESSENTIAL TO MINIMIZING ADVERSE EFFECTS FROM HIGH INTENSITY RADIATED FIELDS (HIRF).**

f) Remove safety clip and close TAWS (G5) (Left Bus) circuit breaker on the avionics circuit breaker panel.

g) Test the EGPWS as per Para 3.A.2).

B. Removal and Installation of EGPWS (MK VI) – (TAWS) Mounting Tray

1) General

These procedures are for the removal and installation of the EGPWS (MK VI) – (TAWS) mounting tray.

2) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Gain access to the forward avionic shelves through the forward baggage compartment.

c) Loosen retaining screw clamp from the front lip of the transceiver and slide the unit out of the tray mount.

d) Remove the four screws that secure the mounting tray to the bottom of the avionics rack.

e) Remove mounting tray from aircraft.

3) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Align the tray to the mounting holes on the instrument panel.

c) Install the four screws that secure the mounting tray to the avionics rack.

5. Troubleshooting

A. EGPWS (MK VI) – (TAWS)

1) If the EGPWS (MK VI) – (TAWS) is deemed inoperative check the following:

a) Ensure TAWS (G5) circuit breaker is closed.

b) Check that the unit is securely installed and the retaining mechanism is securely tightened.

c) Visually inspect EGPWS connector for damaged wires or contacts.

d) Ensure that the Configuration Module has been programmed with the correct aircraft specific values as per Para 3.D.2).

e) If troubleshooting at the wire level is required refer to the applicable wiring diagrams in Wiring Manual PSM 1-64-2W.

f) If repair or service is required, send to appropriate rated avionics repair facility.

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